MIL-S-19500/66B(NAVY)
18 DECEMBER 1968
SUPERSEDING
MIL-S-19500/66A(NAVY)
26 June 1959

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM

TYPE 2N422

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a germanium PNP transistor and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Physical dimensions. See figure 1 (TO-5).

1.3 Maximum ratings.

P_T 1/ T_A = 25° C	V_{CB}	V_{EB}	T_{stg}
mW	Vdc	Vdc	°C
150	-35	-12	-65 to +85

1/ Derate linearly 2.5 mW/° C for T_A > 25° C.

1.4 Primary electrical characteristics.

Limits	h_{fe} V_{CE} = -35 Vdc I_E = 0	C_{obo}	h_{ie}	h_{oe}	f_{hfb}	NF
		pf	ohms	μohms	MHz	db
Min	30	---	18	---	1.0	---
Max	120	60	45	1.0	---	6.5

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The transistor shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin
(b) Manufacturer's designating symbol

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			10				
Emitter to base cutoff current	3061	Bias cond. D V_{EB} = -12 Vdc		I_{EBO}	--	12	µAdc

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TABLE I. Group A inspection (Continued)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2 (cont'd.)							
Collector to base cutoff current	3036	Bias cond. D V_{CB} = -35 Vdc		I_{CBO}	--	20	µAdc
Small signal open circuit output admittance	3216	V_{CE} = -6 Vdc f = 1 kHz I_C = -1 mAdc		h_{oe}	--	1.0	µmho
Small signal short circuit input impedance	3201	V_{CE} = -6 Vdc f = 1 kHz I_C = -1 mAdc		h_{ie}	18	45	ohms
Small signal short-circuit forward-current transfer ratio	3206	V_{CB} = -6 Vdc I_E = 0		h_{fe}	30	120	---
Noise figure	3246	R_L = 100K V_{CB} = -2.5 Vdc I_E = 0.5 ma		NF	--	6.5	db
Subgroup 3			10				
Small signal short-circuit forward-current transfer-ratio cutoff frequency	3301	V_{CB} = -6 Vdc I_E = 1 mAdc		f_{hfb}	1	---	MHz
Collector to base cutoff current	3036	Bias cond. D V_{CB} = -12 Vdc		I_{CBO}	--	20	µAdc
Open circuit output capacitance	3236	f = 1.0 MHz		C_{obo}	--	60	pf

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			20				
Solderability	2026			---	--	--	---
Thermal shock (temperature cycling)	1051	Test cond. A		---	--	--	---
Thermal shock (glass strain)	1056	Test cond. A		---	--	--	---
Moisture resistance	1021			---	--	--	---
End points:							
Collector to base cutoff current	3036	Bias cond. D V_{CB} = -35 Vdc		I_{CBO}	--	25	µAdc

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TABLE II. Group B inspection (Continued)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2 (cont'd.)							
Small signal short circuit forward-current transfer ratio	3206	V_{CB} = -6 Vdc I_E = 1 mA		h_{fe}	24	144	---
Subgroup 3			20				
Shock	2016	Nonoperating; 15,000 G, approx. 0.5 msec; 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1.		---	--	--	---
Vibration fatigue	2046	Nonoperating		---	--	--	---
Constant acceleration	2006	20,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1.		---	--	--	---
End points:							
(Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test cond. E		---	--	--	---
Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	--	--	---
End points:							
(Same as subgroup 2)							

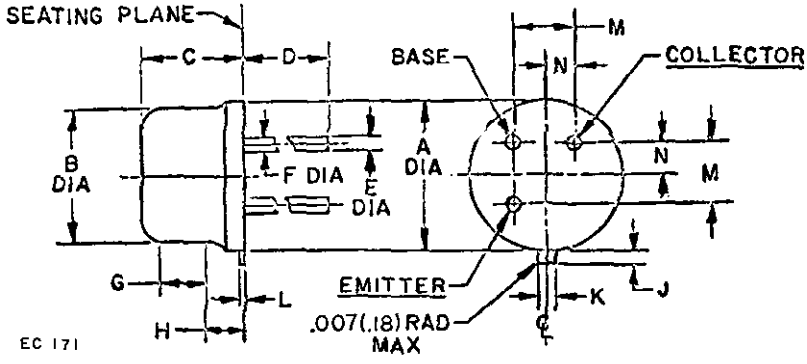
TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = 85° C					
Subgroup 2			λ = 10				
Steady state operation life	1026	V_{CB} = -35 Vdc I_E = 0					
End points: (Subgroups 1 and 2)							

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TABLE III. Group C inspection (Continued)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2 (cont'd.)							
Collector to base cutoff current	3036	Bias cond. D V_{CB} = -35 Vdc		I_{CBO}	--	25	μAdc
Small signal short circuit forward-current transfer ratio	3206	V_{CB} = -6 Vdc I_E = 1 mA		h_{fe}	24	144	---

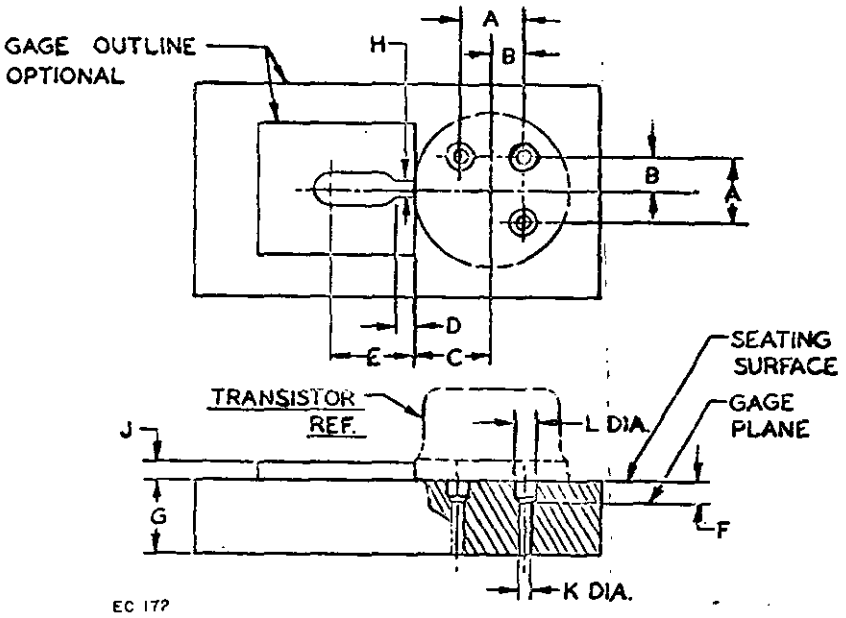


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	0.335	0.370	8.51	9.40	
B	.275	.335	6.98	8.51	
C	.200	.260	5.08	6.60	
D	1.500	--	38.10	--	9
E	0.0169	.0172	0.41	0.436	2, 9
F	.016	.019	.41	.48	3, 9
J	.029	.045	.74	1.14	8
K	.029	.033	.736	0.838	
M	0.1414 Nom		3.59 Nom		6
N	.0707 Nom		1.80 Nom		6

NOTES:
1. Metric equivalents (to the nearest 0.01 mm) in table and parentheses are shown for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond 0.250 (6.35) from the seating plane.
3. Measured in the zone 0.050(1.27) and 0.250 (6.35) from the seating plane.
4. Variations on dimension B in this zone shall not exceed 0.010 (0.25).
5. Outline in this zone is not controlled.
6. When measured in a gaging plane $0.054^{+0.001}_{-0.000}$ ($1.37^{+0.03}_{-0.00}$) below the seating plane of the transistor, maximum diameter leads shall be within 0.007 (0.18) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows preferred measured method.
7. The collector shall be internally connected to the case.
8. Measured from the maximum diameter of the actual device.
9. All three leads.

FIGURE 1. Physical dimensions of transistor, type 2N422 (TO-5).

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DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	0.1409	0.1419	3.58	3.60
B	.0702	.0712	1.78	1.81
C	.182	.199	4.62	5.05
D	.009	.011	0.23	0.28
E	0.125 Nom		3.18 Nom	
F	0.054	0.055	1.37	1.40
G	.372	.378	9.45	9.60
H	.0350	.0355	0.89	0.90
J	0.150 Nom		3.81 Nom	
K	0.0325	0.0335	0.83	0.85
L	.0595	.0605	1.51	1.54

NOTES:
1. The following gaging procedure shall be used: The use of a pin straightener prior to insertion in the gage is permissible. The device being measured shall be inserted until its seating plane is 0.125 ±0.010 (3.18 ±0.25) from the seating surface of the gage. A spacer may be used to obtain the 0.125 (3.18) distance from the gage seat prior to force application. A force of 8 oz. ±0.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.
2. The location of the tab locator, within the limits of dimension C, will be determined by the tab and flange dimension of the device being checked.
3. Metric equivalents in table and parentheses are shown for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Gage for lead and tab location for transistor, type 2N422.

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5. PREPARATION FOR DELIVERY

5.1 *Preparation for delivery shall be in accordance with MIL-S-19500.*

6. NOTES

6.1 Notes. The notes specified in *MIL-S-19500 are applicable to this specification.*

Review activities:
Navy - SH
User activities:
Navy - AS, CG, MC, OS

Preparing activity:
Navy - EC
(Project 5961-N211)

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